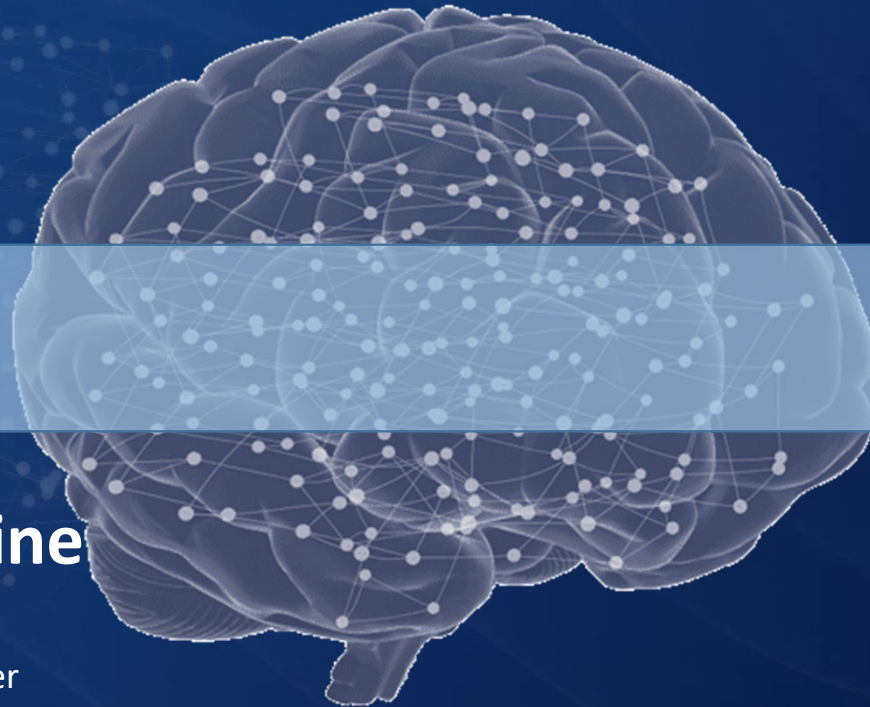


Strong Artificial Intelligence

L Brain

Thinking Machine



Prof. Dr. Dr. h. c. Robert Grebner

Technical University of Applied Sciences, Würzburg, 2026-07-03

Repetition „Motion“

- What is motion?
- Which two categorical motion forms do you know?
- How would you define left and right with basic concepts?
- What is the left-right-paradigm?

Workshop „Brain“

- With all your knowledge you got from this study course, how would you design the architecture of a digital brain?
- What are the core components and core algorithms of your brain machine?
- Why is it so hard, to make an artificial copy of the brain?
- What are the limitations of modern AI approaches compared to our human brain?
- What theories do we need to model a proper brain?

Brain

1. Humans & Machines The Crowning Glory of Creation and Their Creations
2. Thinking Theory Source and Sink of Information and Action
3. Brain Architecture The Thought Machine
4. Brain Screens including Mind's Eye
5. Thought Window Defined Information Exchange
6. AI Spots Machine Learning
7. *Conclusion* Strong Artificial Intelligence



1 Humans and Machines

Crowning Glory of Creation and their Creations

1. Humans & Machines

Humans

What are Humans?

Working Individuals.

What is a Humans Basic Instinct?

Simplify Work.

What are Humans Basically Able to do?

Act & Think

Are Humans Intelligent?

We Have Excellent Sensors, Brains and Musculoskeletal Systems.

1. Humans & Machines Work

What is Work?

Things that have to be done.

What Types of Work Exist?

Physical & Mental

Chances of Work?

Division of Labor.

What Makes Work so Special?

It Connects People and Machines.

1. Humans & Machines Work

What is a Machine?

A Work Performer.

What Brings Humans and Machines Together?

Work Systems.

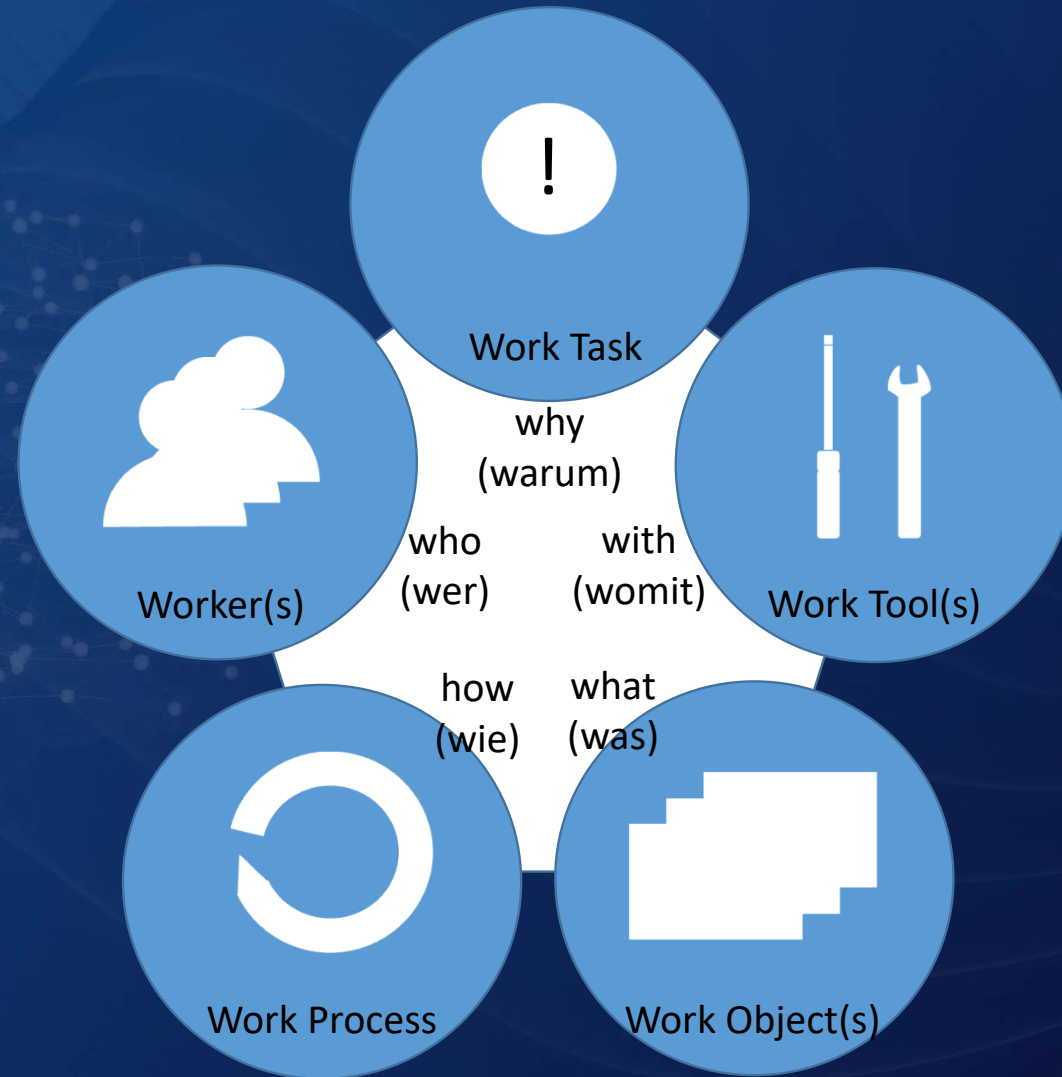
What Kinds of Machines exist?

Mechanic & Non-Mechanic

Which Components Has a Work System?

Task/Purpose Description,
Work Group, Tools, Work
Subject/Piece, Work Process.

1. Humans & Machines Work System



DUDEN

Intelligenz, die

Substantiv, feminin

Fähigkeit [des Menschen], **abstrakt** und **vernünftig zu denken**
und daraus **zweckvolles Handeln** abzuleiten

Ability [of human beings], to **think**
abstract as well as **reasonable** and
derive **purposeful acting** out of that



2 Thinking Theory

Source and Sink of Information and Action

2. Thinking Theory

Disciplines focussing on the brain

Philosophy

research how we see and understand the world (600 B. C.)

Psychology

research interaction behaviour and internal processes (19th cent.)

Cognitive Science

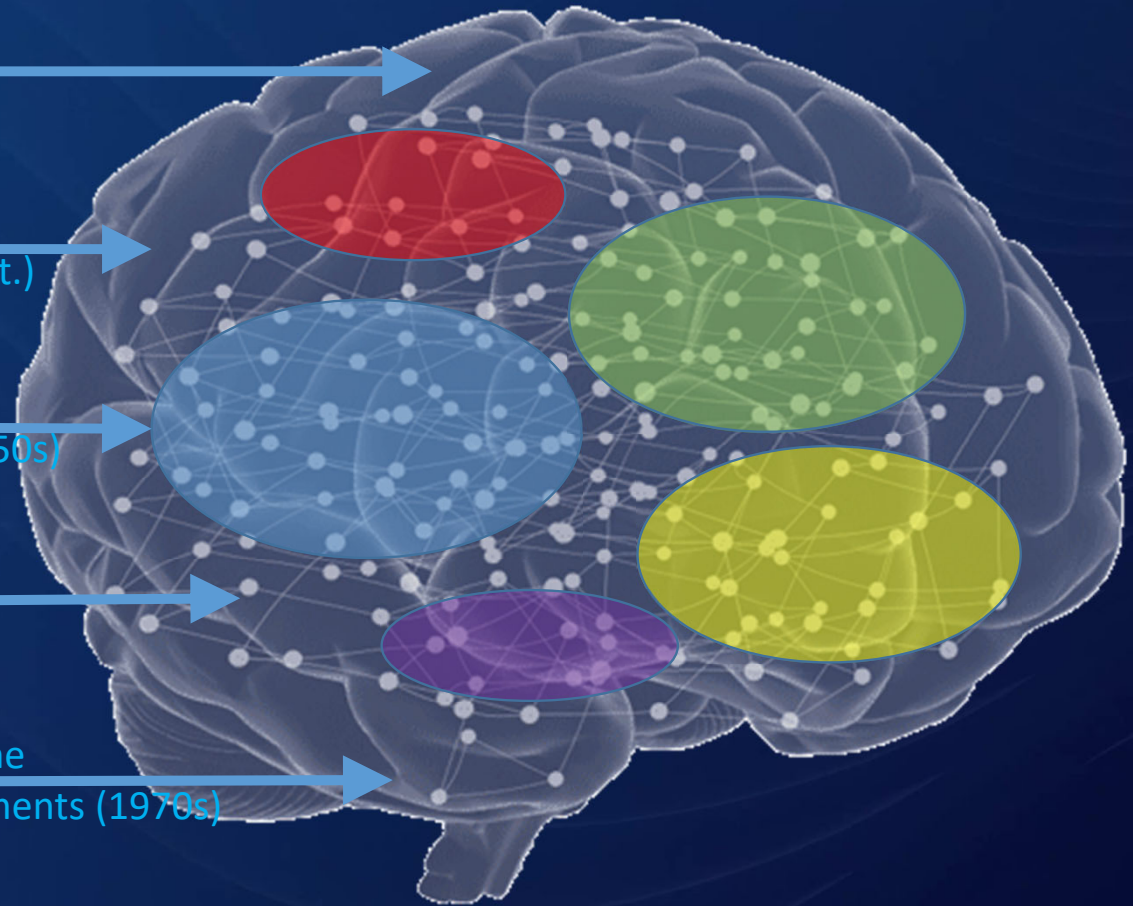
cognition and intelligence models (1950s)

Artificial Intelligence

artificial neural networks (1950s)

Neuroscience

biological research of the interconnected core elements (1970s)



2. Thinking Theory

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Artificial Intelligence

artificial neural networks (1950s)

Neuroscience

biological research of the interconnected core elements (1970s)

automation of information processing
artificial information (since 1937)

Computer
Sciences/
Informatics

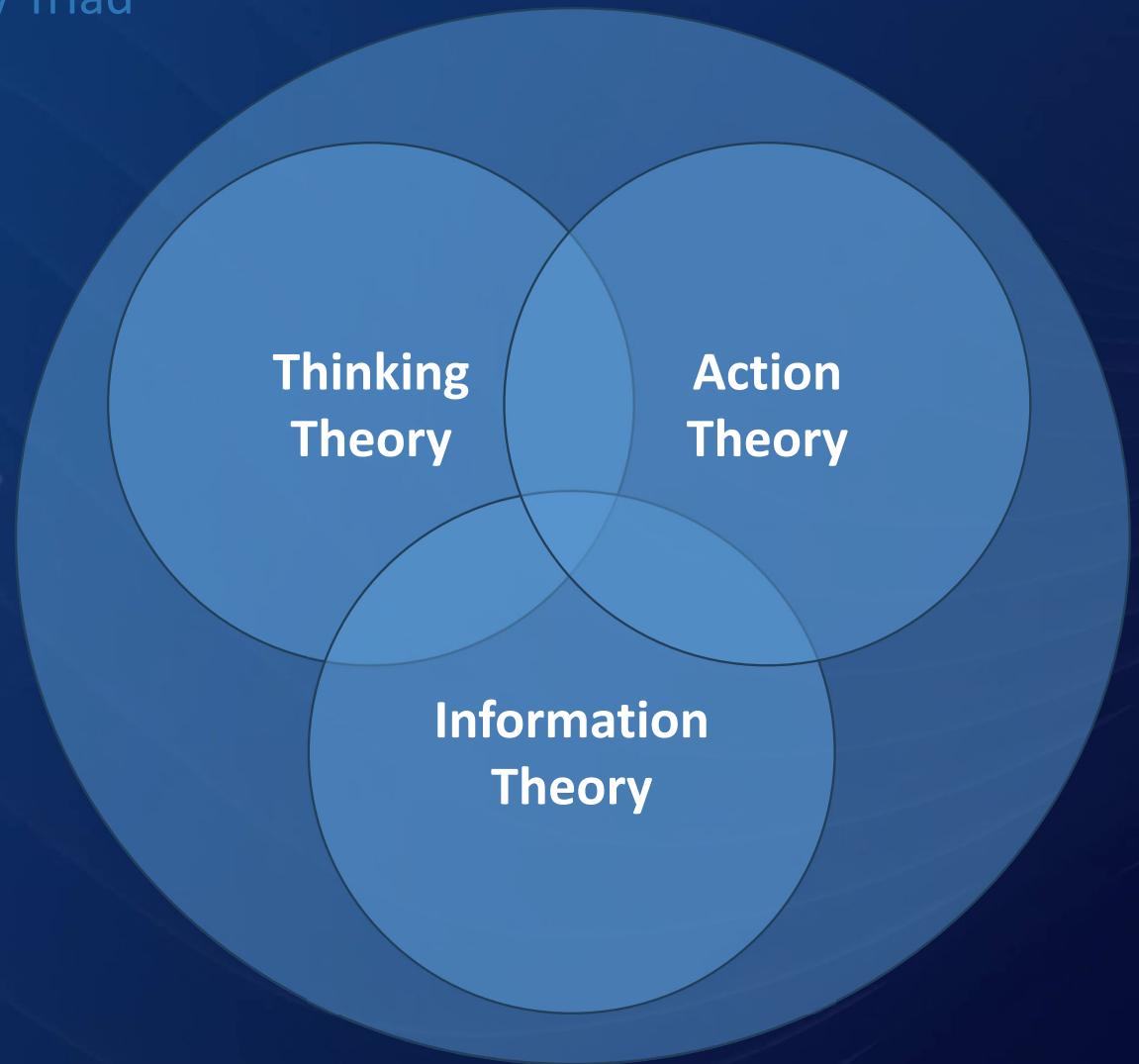


2. Thinking Theory

The Theory Triad

Intelligence

Ability [of human beings], to **think** **abstract** as well as **reasonable** and derive **purposeful** **acting** out of that



2. Thinking Theory

Information Theory: Clarifying Questions

What is Information?

Mental entity.
Representation.

Where is it?

Brain

When is it used?

Thinking.

Why is it (produced)?

Action (from and for).

2. Thinking Theory

Action Theory: Clarifying Questions

What is Action?

Manipulation of
(mental/physical) Entities.

Where is it?

1.) Mental World (Brain).
2.) Physical World based on
an Action Plan.

When is it?

After Identification of a
Need for Action.

Why is it?

Missmatch of Values.

2. Thinking Theory

Status today

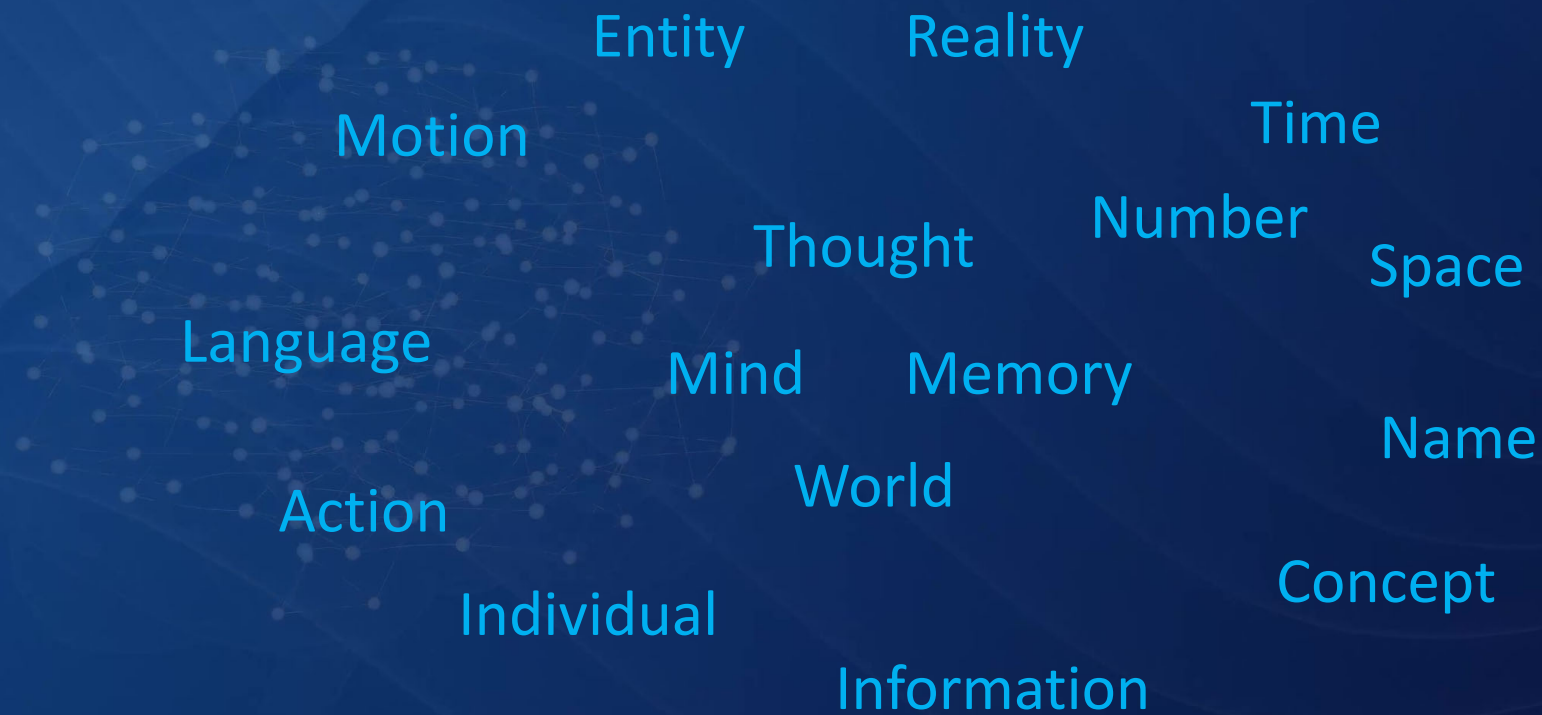
no operationalisable thinking theory

and therefore

no thinking machine until now

2. Thinking Theory

Must Link All Thought Phenomena Together



2. Thinking Theory

The Smallest Thinkable Unit (Elementary Thinking Unit)

Entity

Motion

Thought

Space

Language

Mind

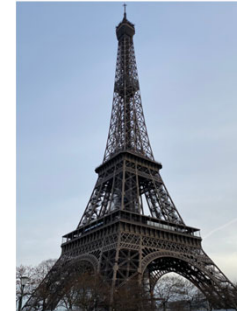
World

Action

Individual

Information

Physical Entities (Matter, Waves)



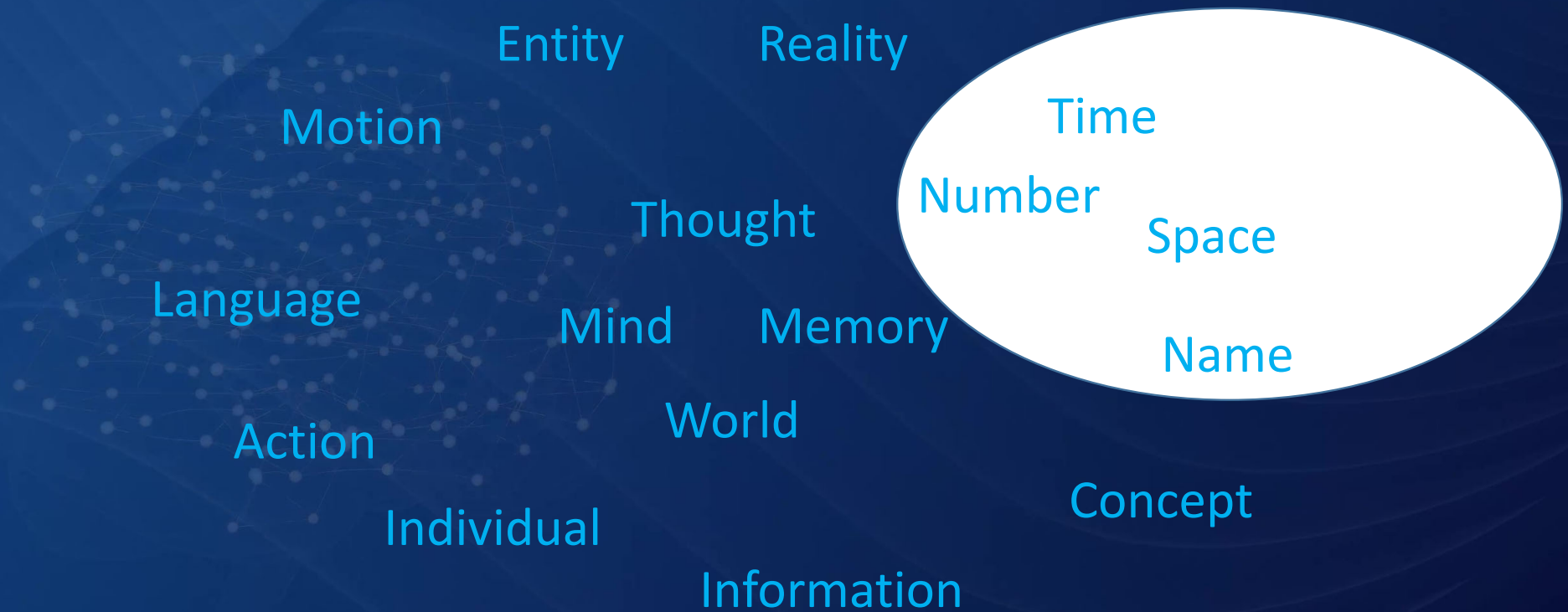
Mental Entities (...)



$\sqrt{x}$

2. Thinking Theory

Elementary Thinkables (Basic Entity Connectors)



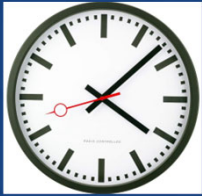
2. Thinking Theory

Meta Thinkable



2. Thinking Theory

Temporal Logic



Entity

Time

exists from <yyyy1-MM1-dd1 HH1:mm1:ss1>

exists to <yyyy2-MM2-dd2 HH2:mm2:ss2>

2. Thinking Theory

Temporal Logic (relative – when is what)



Entity A

any relation between
from

Entity B

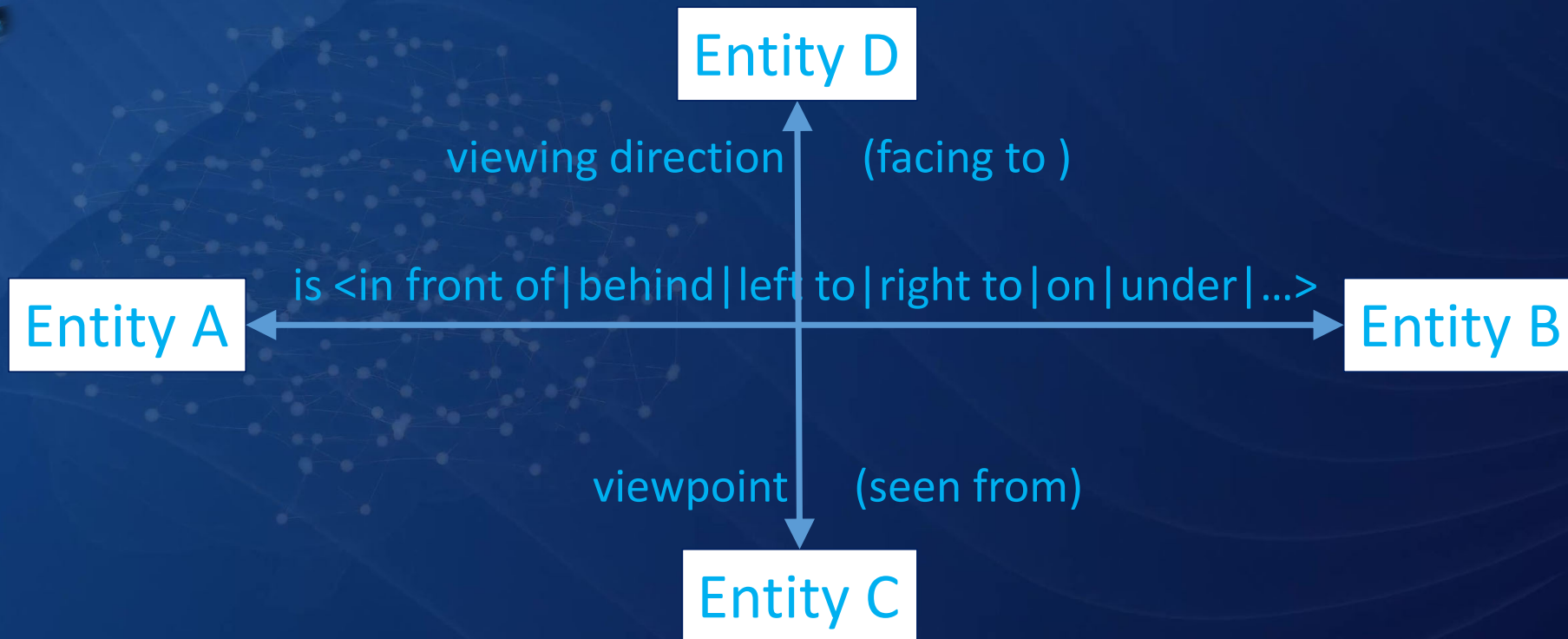
to

<yyyy1-MM1-dd1 HH1:mm1:ss1>

<yyyy2-MM2-dd2 HH2:mm2:ss2>

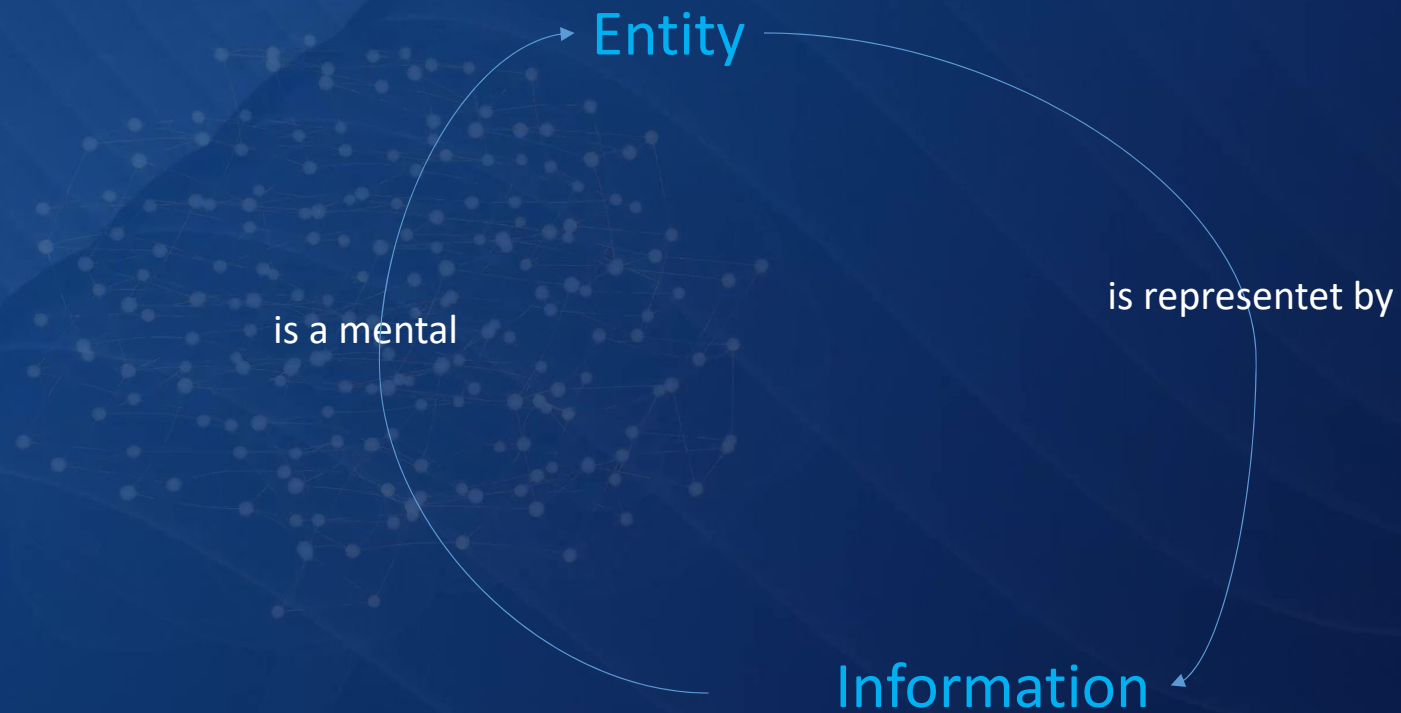
2. Thinking Theory

Spatial Logic (relative – where is what)



2. Thinking Theory

All Basic Elements are Connected



2. Thinking Theory

All Basic Elements are Connected



2. Thinking Theory

Reality Logic (Information Source)

Information A

is expressed by

Individual X

2. Thinking Theory

All Basic Elements are Connected



2. Thinking Theory

Thinking produces concepts by learning

Mind uses concepts (action plans) for speaking and moving

Thinking dedects action necessities

- Reactions to impressions on the base of values
- Imitations
- Curiosity
- ...

2. Thinking Theory

What is Thinking?

Processing of information.

Where is Thining?

Brain.

When is it?

Always.

Why is it?

???

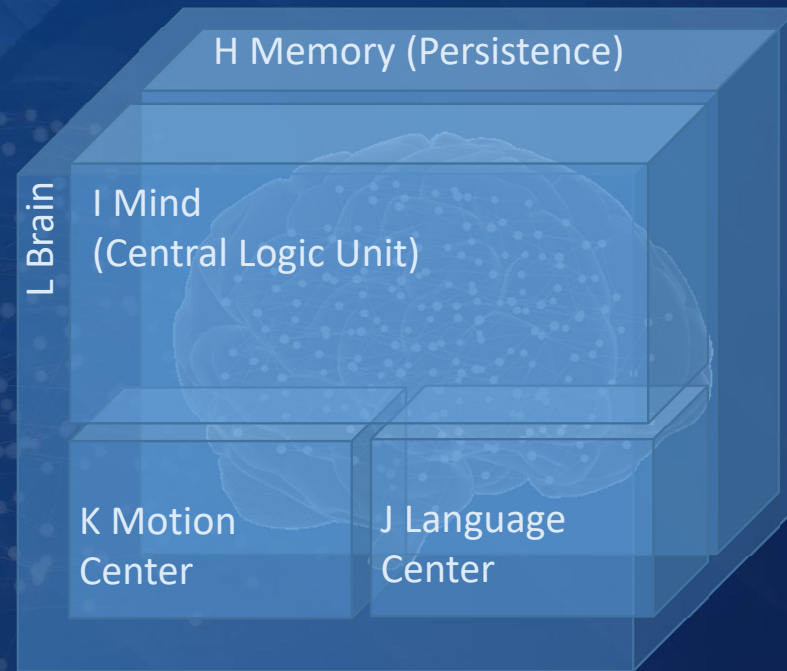


3 Brain Architecture

The Thought Machine

3. Brain Architecture

Brain Sections



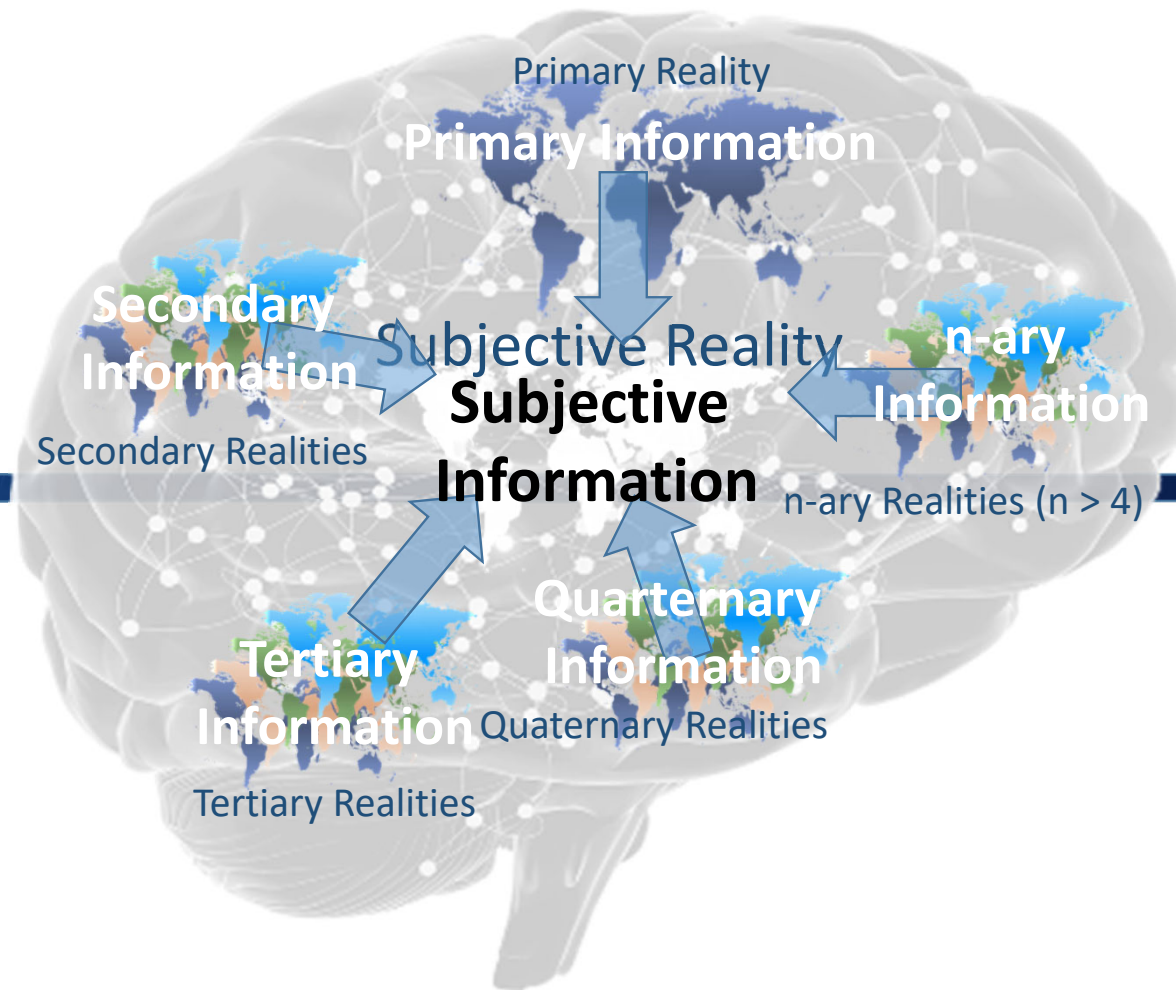
3. Brain Architecture

Memory and Reality Logic (Information Source Levels)

Physical World
(physical entities)

Mental World
(mental entities)

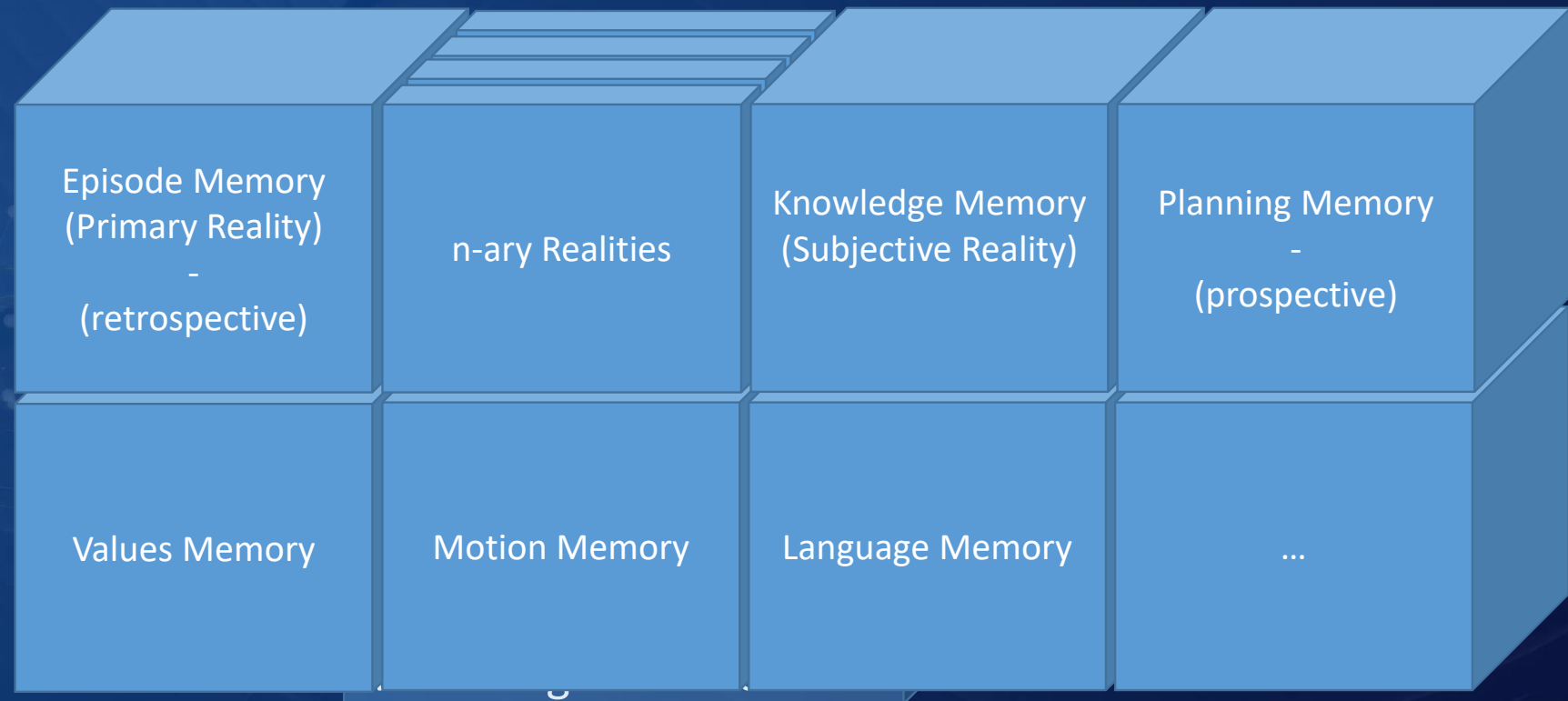
Objective



Reality

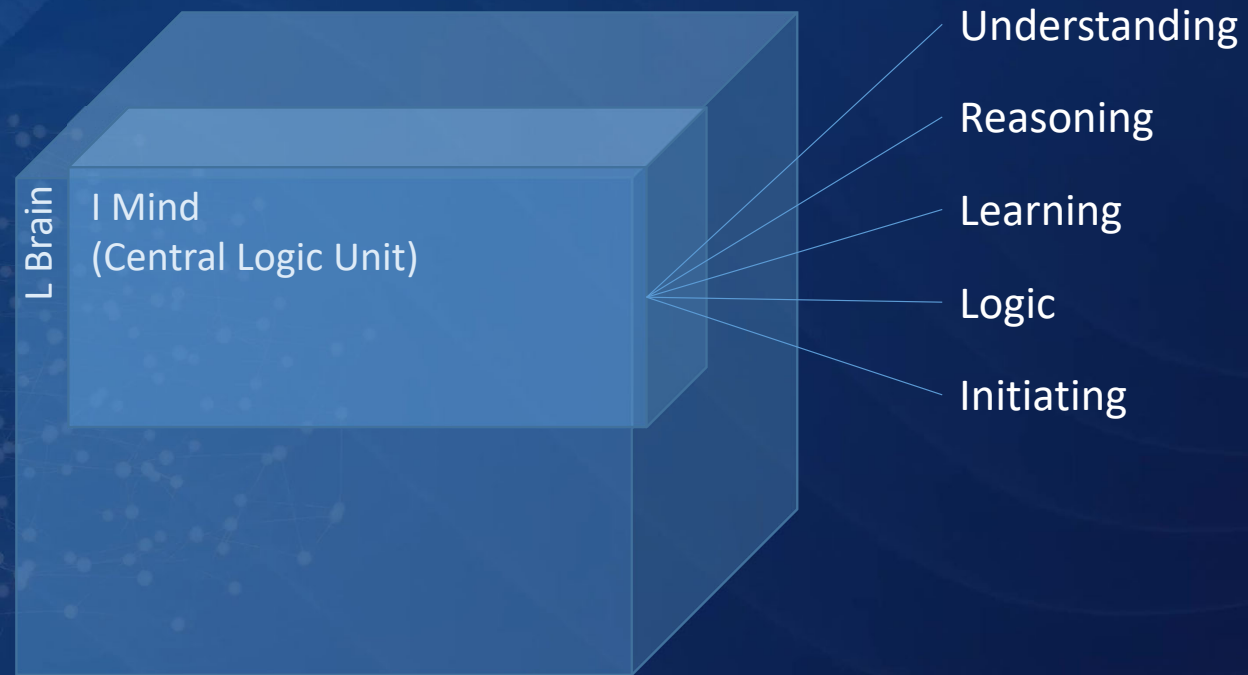
3. Brain Architecture

Memory Sections



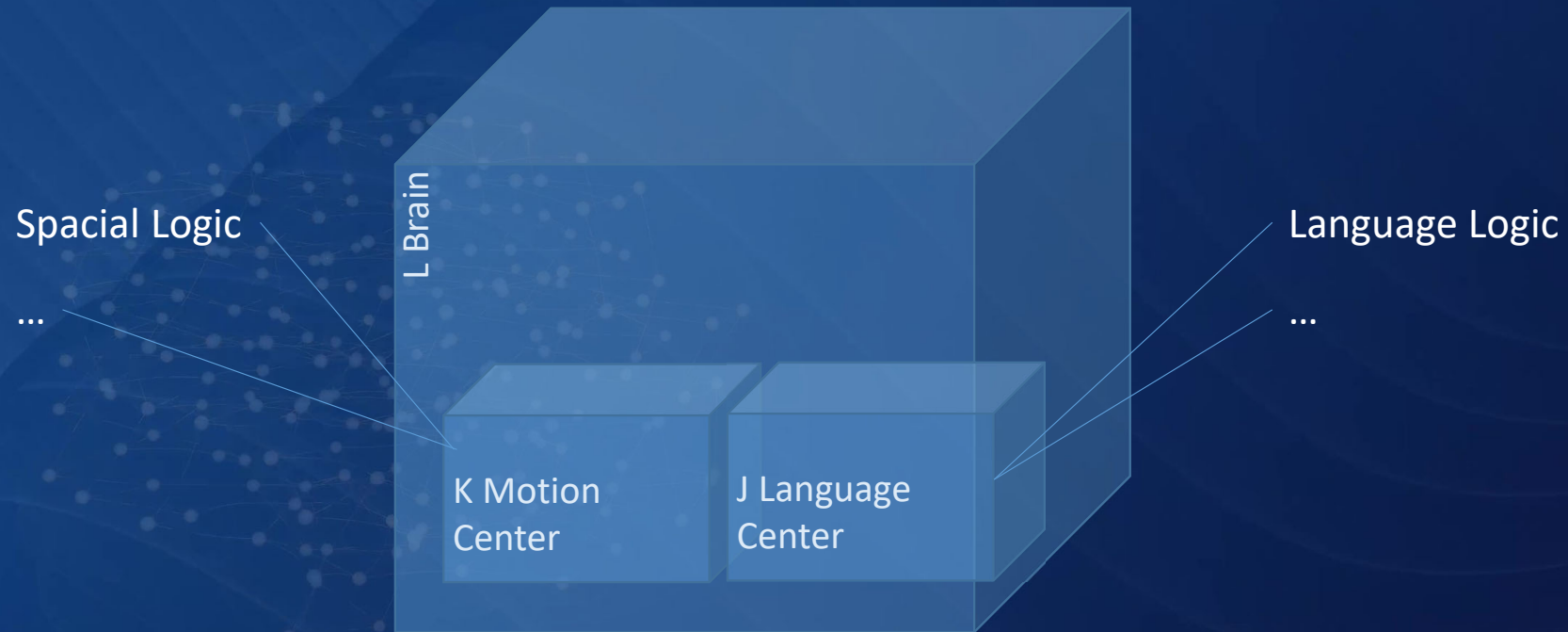
3. Brain Architecture

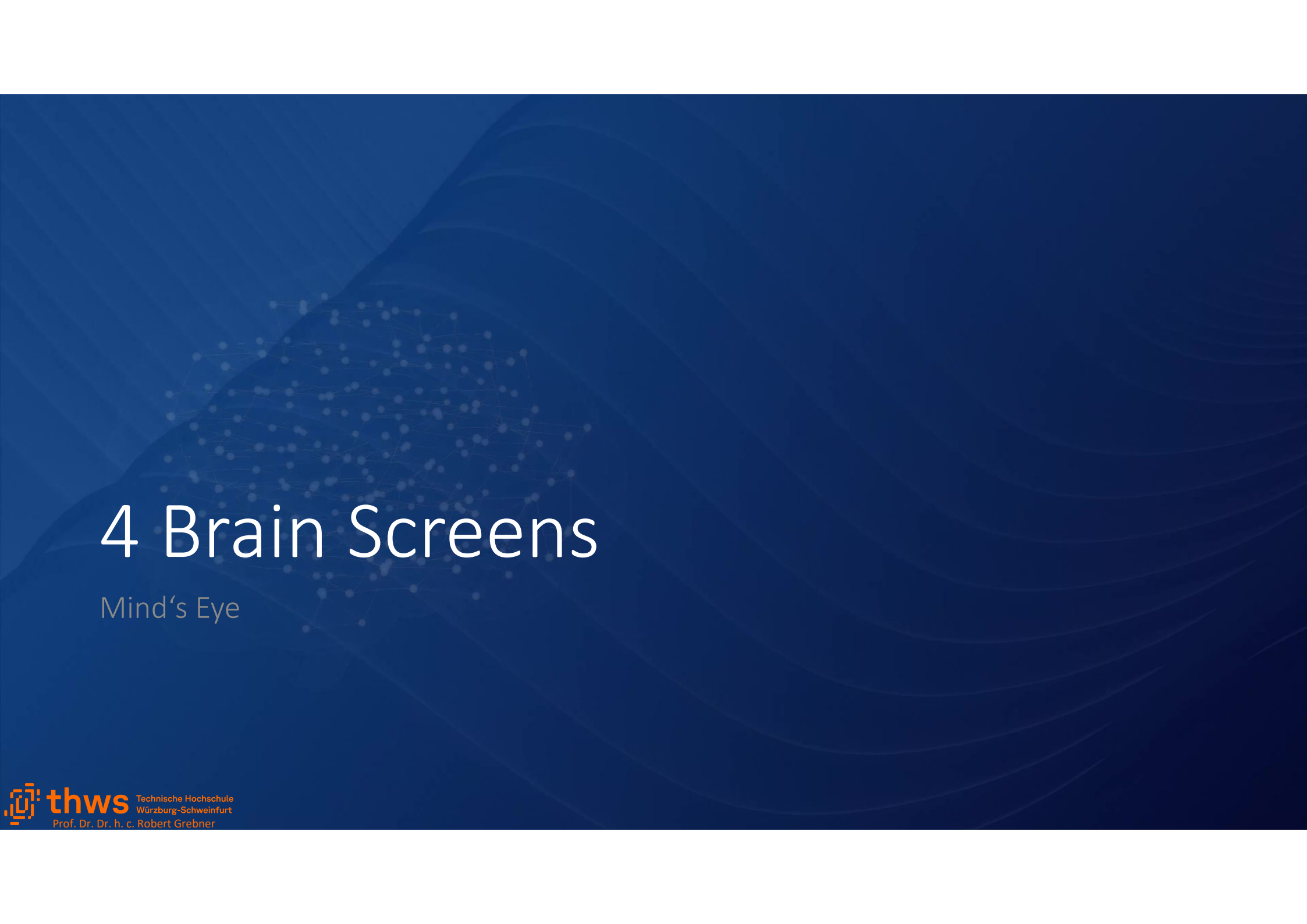
Mind



3. Brain Architecture

Brain Sections

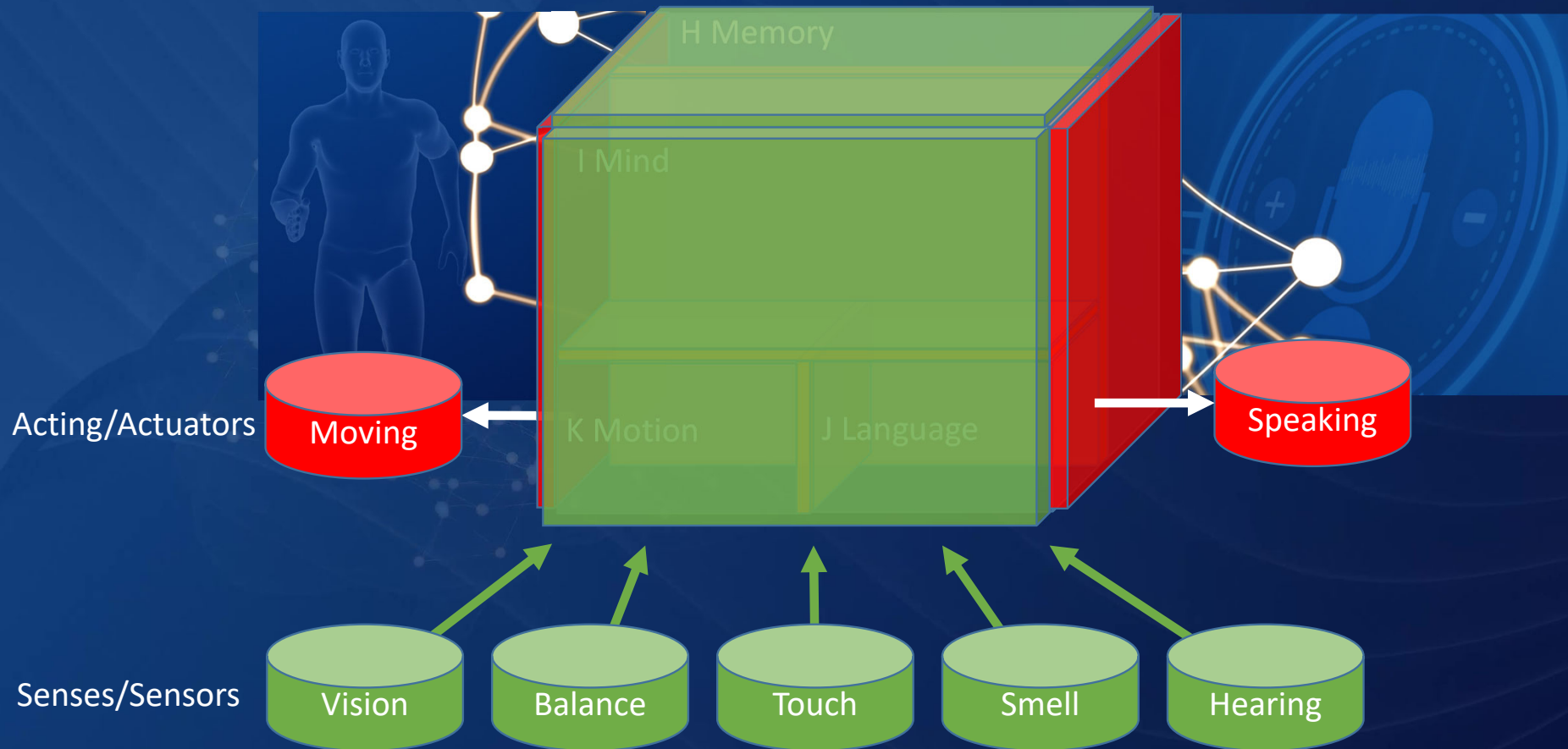




4 Brain Screens

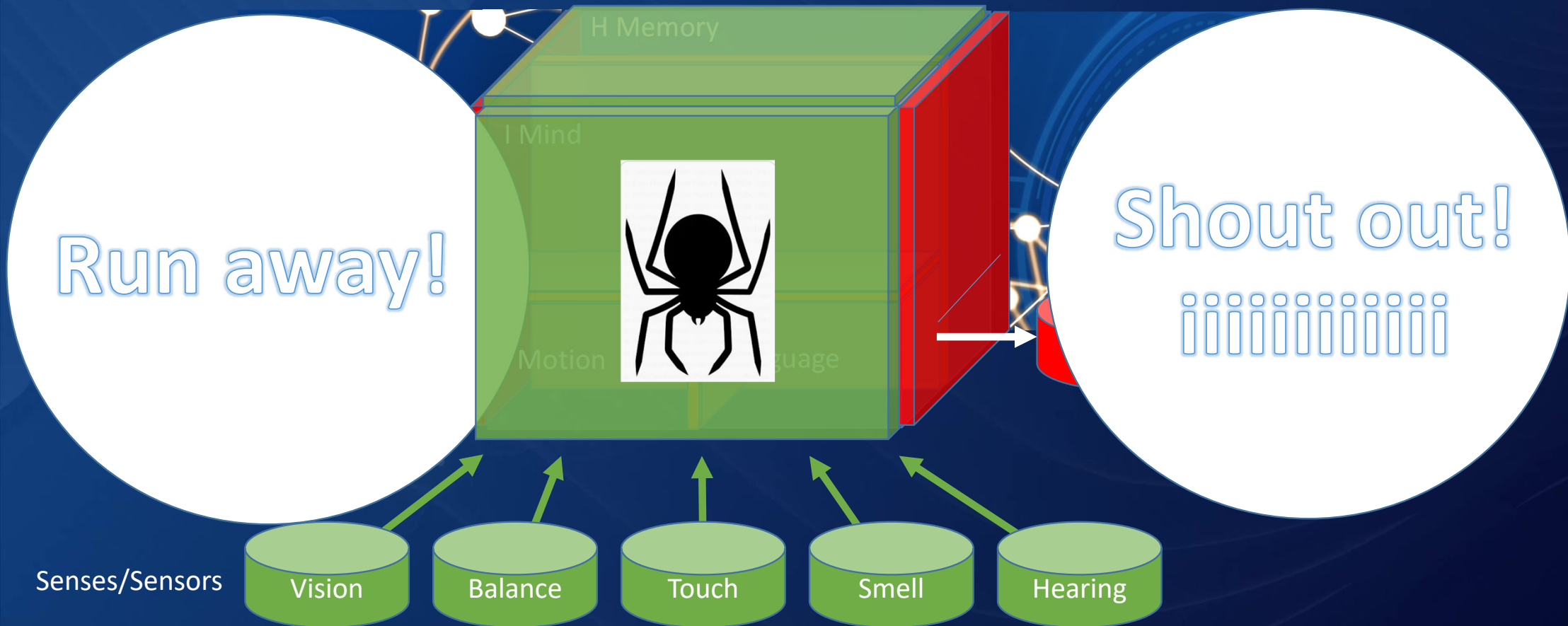
Mind's Eye

4. Brain Screens



4. Brain Screens

Place of Thoughts



4. Brain Screens

- Screens contain transient information, that means it works more as the short term memory of our brain
- The information available can be made visible using a special generic role-entity-based thought language (GREBTL), which is taught in the course Advanced Information Modelling



5 Thought Window

Defined Information Exchange

- Screens are the interfaces among all brain sections
- Screens show thoughts (information networks), which are produced by the connected brain sections
- Every section connected to a screen can see and change the currently displayed thought of that screen (interactive information exchange)
- Example: The motion center and the speech center display the perceived as a thought on the screen connected to the memory; the memory tries to store that information as the primary reality

5. Thought Window

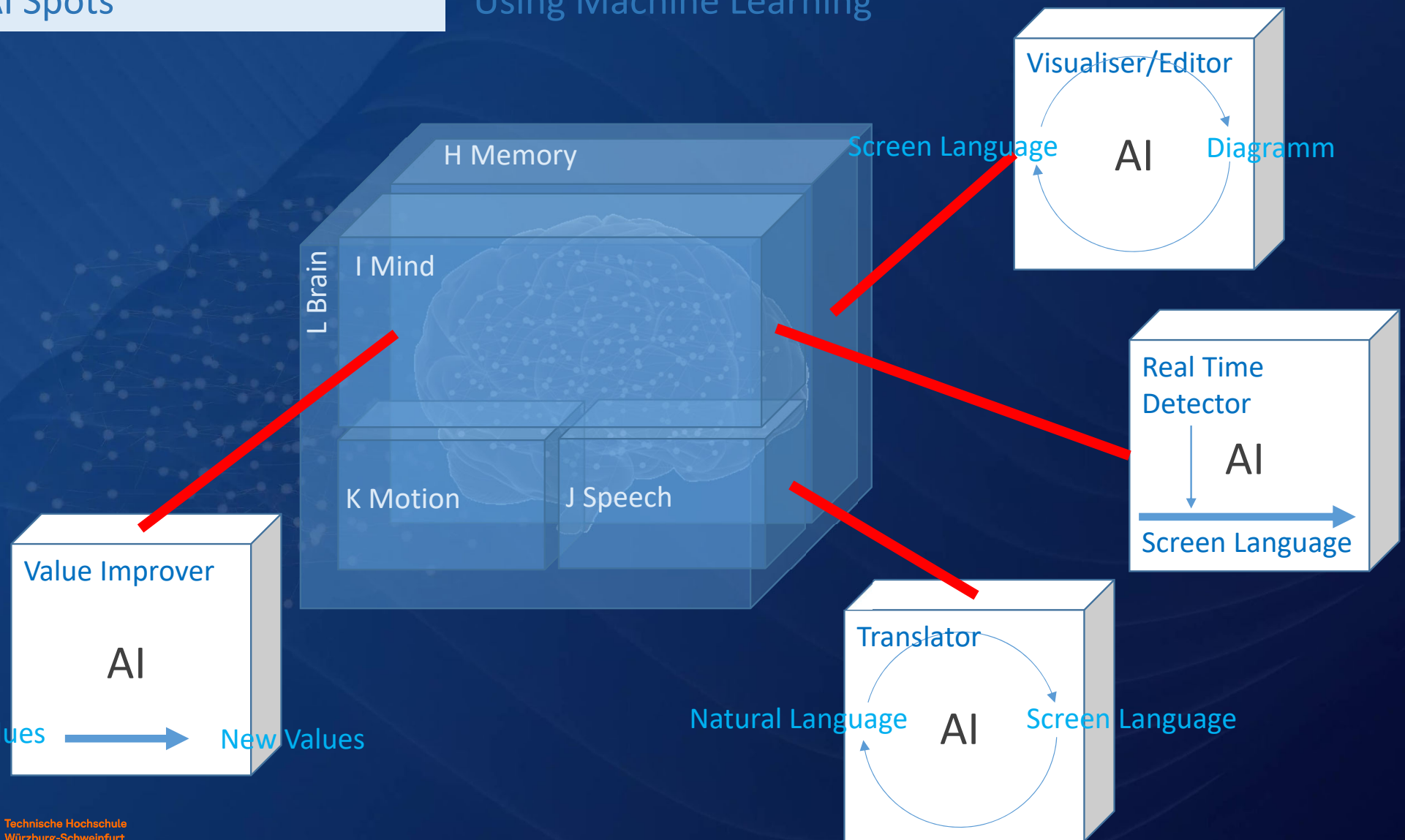
- Thoughts are association networks (and not languages) displayed on a special screen
- Thoughts as information networks can contain a lot of information and are not a sequential channel
- Thoughts or information networks can be changed quickly on a screen by the connected brain component
- The Mind's Eye is that information part of any screen, the brain is just focused on.

6 AI Spots

Machine Learning

6. AI Spots

Using Machine Learning





7 Conclusion

Strong Artificial Intelligence

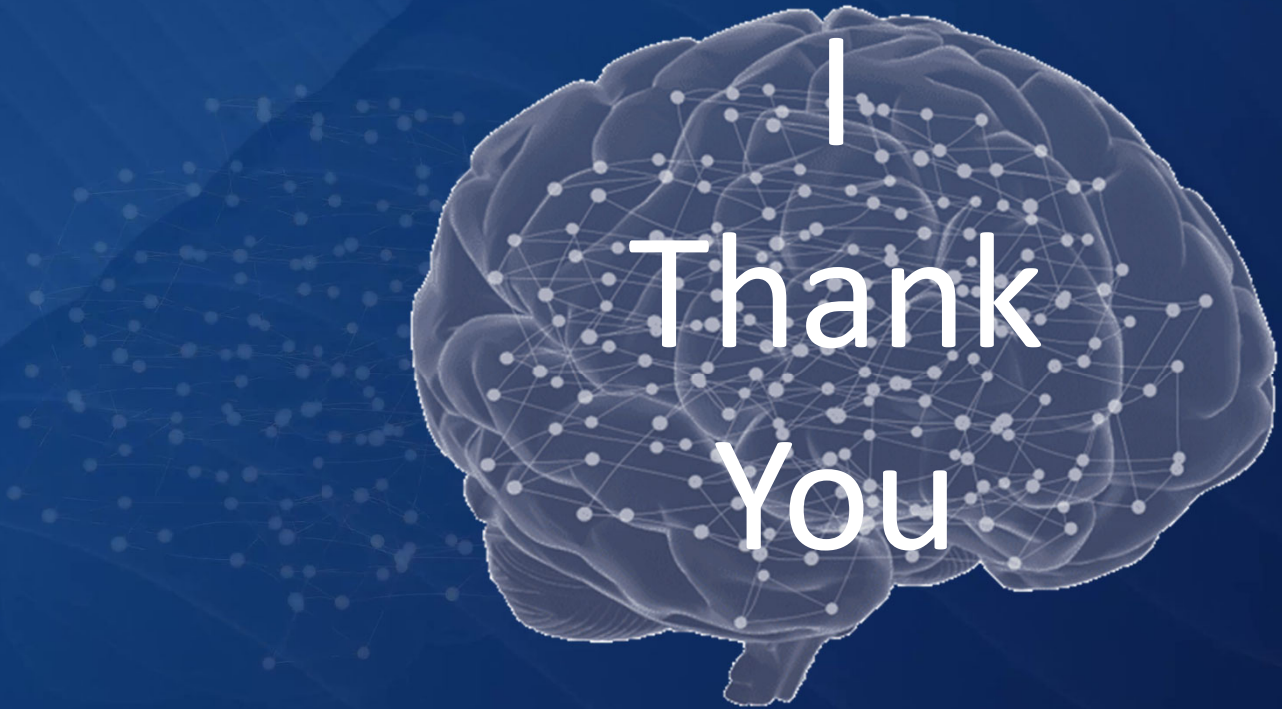
7. Conclusion

- The brain is a machine that processes real information in form of representation networks
- The architecture must not necessarily be complex
- The key probably lies in the different logics that have to be used and bound together to drive that machine

Workshop „Human Brain Project“



- Describe briefly what the Human Brain Project was and what succeeding projects are driven by the European Union. Please make a search/research about that project on public available sources.
- Why was – after ten years research and nearly one billion Euro budget – the „Human Brain Project“ not able to deliver an artificial brain. Find out the reasons.
- How would you set up an project, looking for the implementation for a digital brain.



I
Thank
You